

American Life in Realtime: Benchmark, publicly available person-generated health data for equity in precision health

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Abstract

Person-generated health data (PGHD) from smartphones/wearables are invaluable for precision health, a field promoting health equity through tailored disease prevention, detection, and intervention strategies. However, pervasive convenience sampling in extant PGHD research introduces selection biases that systematically underrepresent disadvantaged groups, limit model generalizability, and risk exacerbating health disparities. Benchmark PGHD (representative, validated, longitudinal, and frequently repeated) are urgently needed to support model equity. To address this fieldwide limitation, we established American Life in Realtime (ALiR), a longitudinal population health study involving PGHD collected from a probability-based, nationally representative cohort using study-provided Fitbits and (as needed) 4G tablets. As a result, ALiR's 1,038 participants are broadly representative across comprehensive sociodemographic, behavioral, and health-related US population norms, overcoming disparities in established convenience samples (e.g. NIH's All of Us; AoU). Only two sources of differential enrollment remained: older age (odds ratio [OR]: 1.27, 99% CI: 1.12–1.45) during consent, lower education (OR: 0.86, 99% CI: 0.79–0.94) during enrollment, though oversampling individuals without bachelor's degrees sufficiently counterbalanced the latter. An illustrative coronavirus disease 2019 classification model—chosen for global significance, known disparities in experience and outcomes, and methodological relevance—trained using ALiR performed equivalently when tested in sample (area under the curve [AUC] = 0.84, 95% CI: 0.79–0.89) and out of sample on AoU (AUC = 0.83, 95% CI: 0.78–0.89) overall, and in historically underserved subgroups (AUC = 0.82–1.0). Conversely, an identically trained classification model using AoU underperformed by 35% out of sample on ALiR (overall AUC = 0.68, 95% CI: 0.61–0.75 vs. AUC = 0.93, 95% CI: 0.91–0.96 in sample), with worse performance in older female and non-White subgroups (by 22–40%). Our results suggest that probability sampling and hardware provisioning enabled cohort inclusivity and generalizable model performance, supporting ALiR's benchmarking potential for equitable recruitment, PGHD collection, and precision health application.

Keywords: person-generated health data, digital health, precision health, health equity, model generalizability

Significance Statement

Data from smartphones/wearables could help personalize healthcare, but existing research underrepresents disadvantaged groups, which risks worsening health disparities. We created American Life in Realtime (ALiR), a wearable-based population health study to address this systematic limitation. By using random, address-based sampling and providing Fitbits and 4G tablets, ALiR removed nearly all participation barriers within disadvantaged groups, resulting in broad representation of US adults compared with existing studies. Further, an illustrative coronavirus disease 2019 detection model—chosen for known disparities, global impact, and methodological comparability—trained using ALiR was equally effective across diverse populations, outperforming an identically trained model using All of Us (NIH's cohort to advance research in underrepresented groups). ALiR thus provides a potential benchmark for equitable recruitment, data collection, and modeling in wearable research.

Competing Interest: M.J., A.B., E.D., and A.M. are affiliated with Evidation Health, Inc. Their contributions were made as part of the NLM-funded grant supporting this study. All other authors declare no competing interests.

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Introduction

Precision health is an emerging field that seeks to characterize and address health disparities through approaches customized to an individual's specific context and needs (1, 2). Current applications, including individualized detection of the novel coronavirus disease 2019 (COVID-19) (3–8), involve AI and machine learning (ML) systems built upon large-scale person-generated health data (PGHD). Derived from routine digital technology engagement (e.g. smartphones and wearables) (9), PGHD can continuously capture everyday real-world social, structural, and environmental exposures, behaviors, and biometrics (10)—factors that account for 60–80% of the modifiable risk of adverse health outcomes (11–14)—providing a valuable opportunity to identify and effectively address unique pathological drivers in marginalized groups.

However, the field lacks “benchmark” PGHD (15, 16), a significant limitation that could exacerbate health disparities in those who may benefit most from precision health advances (17). Benchmarks are standardized datasets that train, evaluate, and validate AI/ML models, ensuring transparency, reproducibility, and generalizability across populations. Without such resources, the field lacks the ability to systematically evaluate model performance, identify biases, and quantify uncertainty, resulting in underperformance in key subgroups and limited real-world applicability. The ideal PGHD benchmark should (i) reflect population diversity; (ii) pair passive data with frequently repeated, validated labels of social, structural, environmental, and health exposures and outcomes; (iii) span longitudinal, and life-course designs; (iv) encompass sufficient data quantity, quality, and diversity for AI/ML applied to individuals, subgroups, or populations; and (v) provide wide accessibility (15, 16). No extant dataset fulfills all five criteria. PGHD cohorts within population studies like NIH's All of Us Research Program (AoU) (18) and UK Biobank (19, 20) systematically underrepresent Black, indigenous, older, lower-educated, or lower-income groups, rely on unstructured labels from claims or electronic health records, and rarely employ repeated, longitudinal instruments (21–25). Most PGHD applications, including COVID-19 detection studies, lack reported demographic distributions, let alone social determinants (3–8). This lack of high-quality, inclusive PGHD, and transparent demographic reporting raises the risk that AI/ML applications may amplify existing disparities, causing underperformance in minority groups (26–28), patient harm (28, 29), and billions in societal costs (30).

COVID-19 highlighted the existence of profound, socially determined healthcare disparities in the United States (31), as racism, poverty, and inaccessible healthcare and social services doubled the morbidity and mortality risk in Black and low-income groups (32). PGHD-based COVID-19 detection thus illustrates both the promise of agile, individualized care that could help circumvent traditional access barriers, and the simultaneous risk of systematic underrepresentation and disparity exacerbation in minoritized groups (3–8).

Whether PGHD inequities stem from methodological selection biases or other barriers remains an open question. Pervasive convenience sampling of easy-to-reach populations (e.g. employers and health systems) and/or “bring-your-own-device” (BYOD) recruitment of self-motivated users afford few participation opportunities for underserved individuals (3–8, 21–25). When opportunities exist, competing time pressure, poor technological literacy, mistrust, privacy/security concerns, or simple disinterest may further selectively erode participation (25, 33–35).

American Life in Realtime (ALiR) was designed as a benchmark for fostering equitable precision health, incorporating wearable PGHD and comprehensive, validated, longitudinal surveys from a nationally representative sample (36). Here we characterize how ALiR's use of probability sampling and study-provided hardware—strategies that reduce methodological participation barriers—impacted cohort inclusivity, differential enrollment, and generalizability of an illustrative COVID-19 classification model compared with data from AoU (a broad convenience cohort designed to advance precision research in underrepresented populations) (18).

Results

Design

Overview

ALiR's design (36) (Fig. 1A; [Online Methods](#)) improved on methodologies of extant population health studies with PGHD collection (18, 20, 37) by incorporating best practices from multiple disciplines such as probability sampling, data benchmarking (15–17), and FAIR (38) (findable, accessible, interoperable, and reusable) standards (Fig. 1B). ALiR's participants were invited from the Understanding America Study (UAS) (39, 40), a nationally representative research cohort. Consenters were given a Fitbit Inspire 2 (Google, LLC/Fitbit Inc.; San Francisco, CA, United States) for study wear (on the wrist) as an incentive. For each individual, at least a year of data collection overlays (i) continuous Fitbit biometrics; (ii) frequent, longitudinal surveys eliciting sociodemographics, social, structural, and environmental exposures, personality, behaviors, and health measured via high-quality, validated instruments; and (iii) geospatially and temporally matched contextual linkages. A custom study app—broadly compatible across devices, operating systems, and firmware—facilitates enrollment, data collection, and long-term engagement through earned points redeemable up to \$126 in annual compensation. ALiR's infrastructure, data, and study app code will be publicly available (anticipated late 2025). ALiR was approved by USC's Social and Behavioral IRB (UP-21-00181) and follows STROBE guidelines.

Population

ALiR participants were randomly sampled from UAS (IRB: UP-14-00148), a probability-based panel representative of US adults (18+). UAS is considered one of the richest sociodemographic data sources (39, 40), generating important, generalizable findings across multidisciplinary domains (41–43). UAS members ($n=9,281$ at the time of enrollment, but actively growing to 20,000 by 2026) are recruited by address-based sampling (14.4% average empanelment rate across recruitment batches) and given a 4G Tablet if they lack Internet, minimizing selection biases (e.g. random digit dialing suffers from diminishing landline use; digital advertising excludes those with access barriers). UAS members answer English/Spanish surveys via mobile/web interface and are compensated (\$40/h). Since inception in 2014, a comprehensive survey core is fielded to the whole panel and repeated biennially (40). Response rates are high (>75% cumulative) and attrition is modest (<8% annually). Sample weights (sex, race/ethnicity, age, education, and geography) match UAS demographic distributions to population norms from the Current Population Survey (CPS) (40, 44).

Study activities

Study activities occurred remotely for at least 1 year (Fig. 2). A 5-min informed consent survey was fielded through the customary UAS electronic interface. During enrollment, consenting individuals were sent

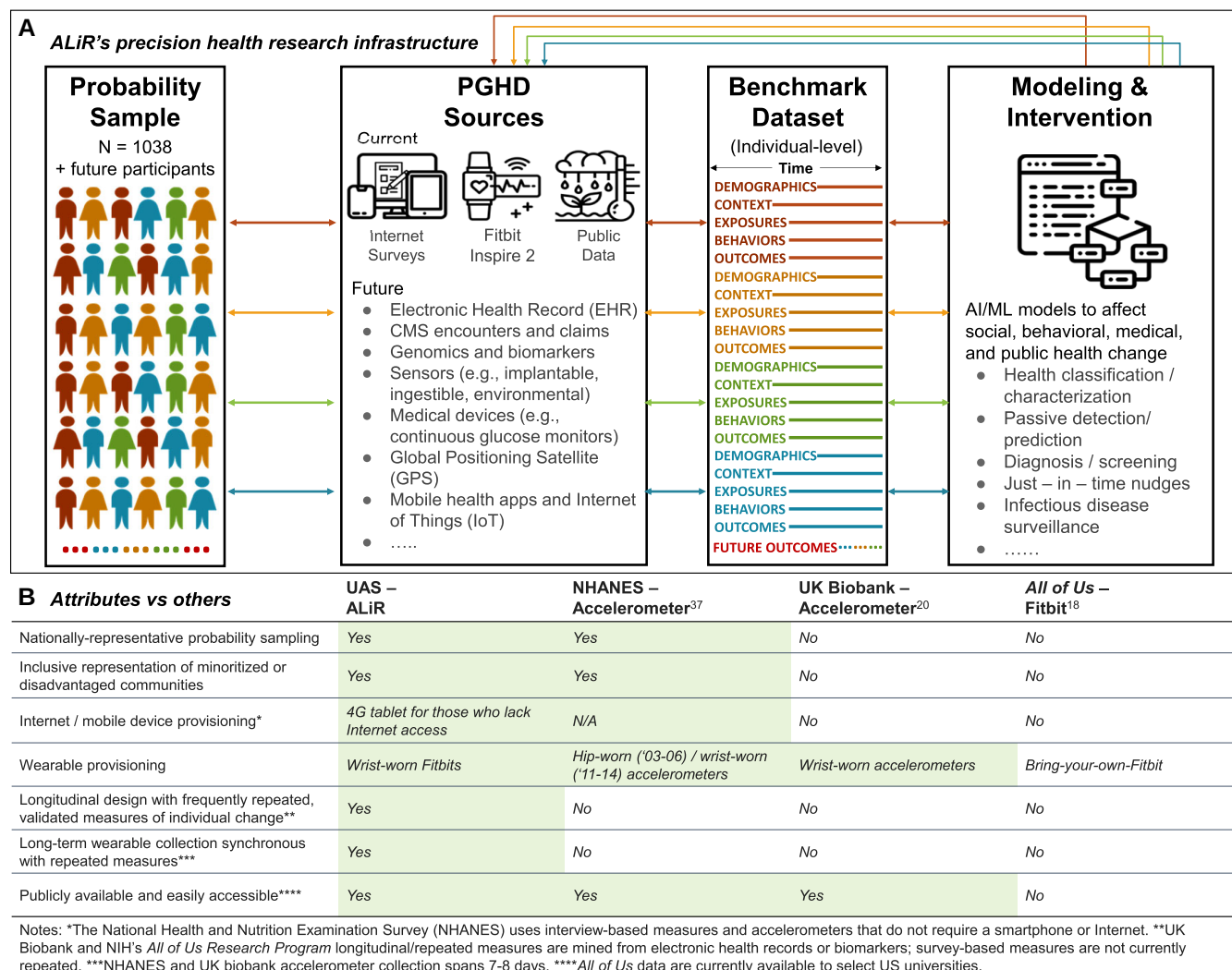


Fig. 1. ALiR design and attributes vs. other population health PGHD cohorts. A) Schematic representation of ALiR infrastructure, including an established, connected cohort, study app with current and future integrations, and benchmark PGHD dataset. B) Comparison of design features across benchmarking and FAIR criteria.

follow-up emails (or phone calls as needed) to confirm current mailing addresses. Fitbits and enrollment instructions were mailed to participants via signature-required, 2-day FedEx. Participants were instructed to download/install the study app from the relevant iOS/Android store, which then facilitated downloading and setting up the Fitbit device and app and permitting/syncing data exchange. Ongoing Fitbit activities involve regularly charging and continuously wearing the provided device, including during sleep. ALiR surveys are short but frequent, fielded every 1–3 days, consist of 10–15 questions, and take 1–2 minutes to complete (Appendix 1). The UAS help-desk is available to assist on demand.

Data collection

Data collection, designed for label validity in supervised learning (36), uses a comprehensive array of frequently repeated consensus measures (e.g. Health and Retirement Study; HRS (45), NIH Phenotypes and Exposures; PhenX (46), Patient Reported Outcomes Measurement Information System; PROMIS (47)) to longitudinally characterize each participant along demographic, socio-economic, personality, behavioral, and health factors (Fig. S1). UAS core data, repeated biennially, comprises 5,000+ variables

across 10+ years and 600+ surveys (40), including (but not limited to) (i) sociodemographics; (ii) all variables from HRS (45); (iii) comprehensive cognitive tests and impairment classifications (48–56); (iv) psychological, well-being, and mental health scales (e.g. CES-D (57)); (v) personality (Big Five (58)); (vi) economic well-being and financial planning; (vii) competencies and literacies; (viii) COVID-related experiences; and (ix) behaviors/perceptions. ALiR-specific surveys (Appendix 1) assess multiple dynamic health-related factors, repeated monthly (or annually where relevant), including (x) social/structural determinants (e.g. food/housing security, racism/discrimination, neighborhood safety, and adverse childhood events); (xi) behaviors (e.g. drugs/alcohol, diet, internet competence, health literacy, health motivations, medical mistrust, vaccine hesitancy, etc.); and (xii) health/well-being (e.g. mental/physical outcomes, infectious disease, healthcare utilization, etc.). (xiii) ALiR Fitbit data include both minute-level sensor data and Fitbit's aggregated metrics (e.g. heart rate, heart rate variability, activity, steps, sleep, temperature, blood oxygenation, currently; others as released by Fitbit, in the future) (59). (xiv) UAS linkages matched spatially/temporally at relevant frequencies include air/water quality, weather, crime, neighborhood characteristics, Centers for Medicare and Medicaid Services claims, Social Security

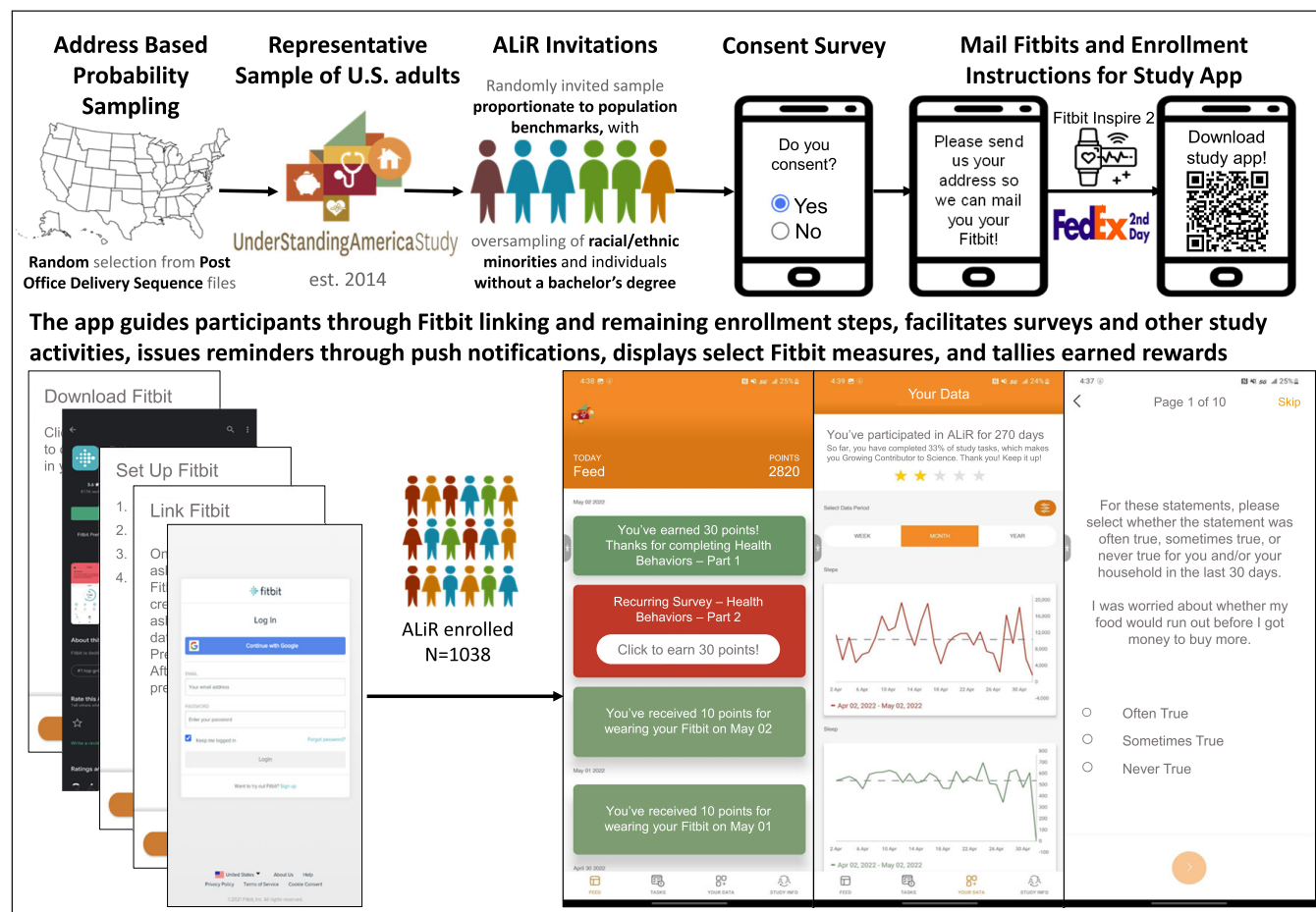


Fig. 2. ALiR participant journey: participant onboarding steps including study app screenshots.

Administration earnings/benefits, and the national death index (40). Finally, (xv) UAS prospective studies are added as relevant, like whole-genomic sequencing and personal wearable-based pollution monitoring (Atmotube).

Enrollment

Sample selection

From 2021 August 3 to 2022 March 1, we randomly invited 2,468 UAS members to ALiR using 16 demographic groups defined by intersections of biological sex, age, race/ethnicity, and education (Table S1). Relative to population benchmarks, non-Whites were oversampled by 40% and noncollege graduates by 5%. We excluded individuals with computers only (7% of UAS; Fitbit required either smartphone or tablet), and individuals needing mobility assist devices (6% of UAS; there were few validated Fitbit measures for this population at the time).

Nonresponse

The consent survey achieved a response rate of 87.1% ($n = 2,150$). There were no major differences in the likelihood of completing the consent survey across demographic groups (Tables S2 and S3).

Consent

Of consent survey respondents, 64.4% ($n = 1,386$) agreed to participate in ALiR (Fig. 3). The most common reason for nonconsent in survey comments—which were only provided by 12% of respondents and may not be generalizable—was already having an “Apple

Watch” or “better” device (23%). Most of these individuals (85%) were White with household incomes above \$60 K. An additional 20% of commenters cited logistics (e.g. work environments that cannot accommodate wearables) or unwillingness to continuously wear a device. Only 16% cited barriers like mobile network coverage or privacy (Table S4).

Enrollment

As of 2022 April 4, 74.9% ($n = 1,038$) of the consenting sample enrolled in ALiR and 25.1% ($n = 348$) were lost to follow-up. Of this latter group, 75.3% ($n = 262$) did not confirm their address and could not be sent their device; 12.1% ($n = 42$) received materials but never enrolled; and 12.6% ($n = 44$) withdrew. Almost all withdrawals (86%; $n = 38$) occurred before enrollment (Table S5). Reasons for withdrawal varied but included repeated unavailability to sign for packages due to work or other commitments, study time commitment, phone/tablet incompatibility with the Fitbit app, and device-related skin irritation.

Barriers and differential enrollment

We used logistic regression and random forests (RFs) to identify observed factors associated with nonconsent and nonenrollment after consenting across combinations of 79 relevant variables (sociodemographics, health, behaviors, personality, skills, cognitive ability, etc.; Online Methods, Appendix S2A). Odds ratio (OR) estimations below refer to the most parsimonious regression (17 variables: biological sex, race/ethnicity, education, age, marital

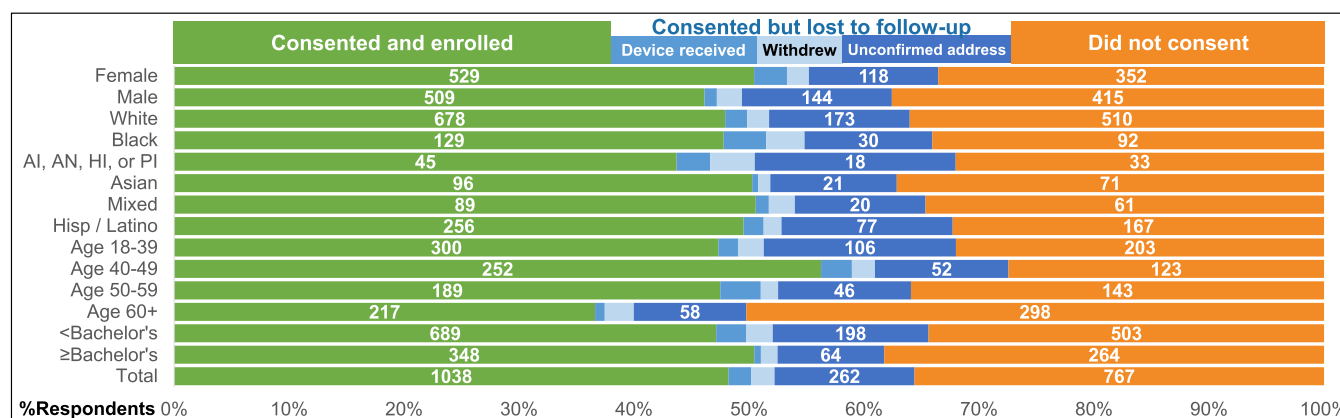


Fig. 3. ALiR enrollment: distribution of enrollment status among consent survey respondents within sociodemographic subgroups: orange = did not consent; shades of blue = consented but lost to follow-up; green = enrolled. Data labels = counts. AI, AN, HI, or PI = American Indian, Alaska Native, Hawaiian, or Pacific Islander.

status, labor status, income, Census region, urbanicity, household size, self-reported health, body mass index (BMI), and number of preexisting chronic conditions). See Figs. S2–S4, Appendix S2B for additional results and sensitivity analyses.

Older age (OR: 1.27, 99% CI: 1.12–1.45) was the only observed historically underrepresented or disparity-related factor consistently associated with nonconsent (i.e. unwillingness to participate) across all models after correction for multiple hypotheses (False Discovery Rate; FDR < 0.05; Fig. S2). Other factors—more commonly associated with higher sociodemographic strata—included smaller household size (OR: 0.90, 99% CI: 0.82–0.98) and lower BMI (OR: 0.76, 99% CI: 0.62–0.92). RFs identified income and nonhousing wealth as barriers with nonlinear interactions. Heterogeneity analysis (60) suggested that participants with the highest total wealth and picture-vocabulary scores were also less likely to consent (both factors are more prevalent within higher sociodemographic strata).

Lower educational attainment (OR: 0.86, 99% CI: 0.79–0.94) was the only observed disparity-related variable consistently associated with nonenrollment after consenting (i.e. barriers beyond willingness to participate; Fig. S3). Heterogeneity analysis (Fig. S5) suggested that individuals with the lowest educational attainment (<9 years) were least likely to enroll.

Systematic univariate testing for nonenrollment across all 5,000+ UAS variables did not yield any additional factors not related to those in our main models (e.g. age; Tables S6 and S7).

ALiR's representation vs. population norms

Overall, ALiR's unweighted (i.e. raw) cohort resembled the US adult population when compared with a holistic set of population norms (see Online Methods), including demographics, socioeconomic, education, personality, behaviors, and health (Fig. 4, Tables S8 and S9). The most statistically significant differences relative to the population were expected given the sampling strategy and/or the composition of the parent UAS cohort. Racial/ethnic minorities (oversampled by 40%) are overrepresented by 42% in ALiR (54% vs. 38% in the population; Fig. 4A), while White individuals are underrepresented by 26% (46 vs. 62%). Since racial/ethnic minorities are more likely to be lower on the income distribution, ALiR overrepresents those with annual household incomes <\$30,000 by 16% (22 vs. 19%) and underrepresents those with ≥\$100,000 by 19% (26 vs. 32%; Fig. 4B). ALiR's geographic composition mirrors UAS, where the Northeast and South are

underrepresented, while West is overrepresented due to large (deliberate) California and Los Angeles County subsamples (Fig. 4D). Finally, individuals with functional limitations are slightly underrepresented due to the exclusion of those needing mobility assist devices (Fig. 4I).

Unweighted ALiR's age discrepancy was not anticipated. The fraction of individuals aged 60+ in ALiR is 32% below population norms (21 vs. 31%). Consequently, the fraction of retirees is 40% lower (12 vs. 20%).

While our sampling strategy focused on achieving sociodemographic representation, ALiR's unweighted sample is also representative of US adults across personality (Fig. 4E), skills (Table S8), and several dimensions of health (Fig. 4F–J, Table S9). Those with digital access barriers are also included, as 77.2% and 2.3% did not have a wearable or Internet access prior to UAS-ALiR provisioned hardware, resembling respective norms (Fig. 4C).

Application of sample weights—generated to align ALiR's distributions of basic demographics to the adult US population (biological sex, race/ethnicity, age, education, and geography; Online Methods)—corrected for most observed discrepancies, including variables not used in weighting. Three notable discrepancies remained. First, the weights improve but do not fully correct for the underrepresentation of retired individuals. Second, weights do not correct for the overrepresentation of unemployed, on leave, and disabled individuals. Third, individuals with diagnosed hypertension are underrepresented while those with diabetes are slightly overrepresented (61).

ALiR's representation vs. convenience-based cohorts

To evaluate improvement in representation, we compared ALiR's unweighted sample to two comparison cohorts that isolate benefits from different aspects of our approach. (i) UAS-BYOD, a sample of wearable owners within the UAS probability sample, is used to assess the impact of hardware provisioning vs. BYOD (Fig. 4, Tables S10 and S11). (ii) AoU's BYOD, a clinical convenience sample of Fitbit owners, is used to assess the combined benefits of probability sampling and hardware provisioning (Table S12).

Compared with population norms, UAS-BYOD ($n = 1,631$) significantly underrepresented individuals who are: (i) male by 32%; (ii) Black by 48%; (iii) Asian by 29%; (iv) American Indian,

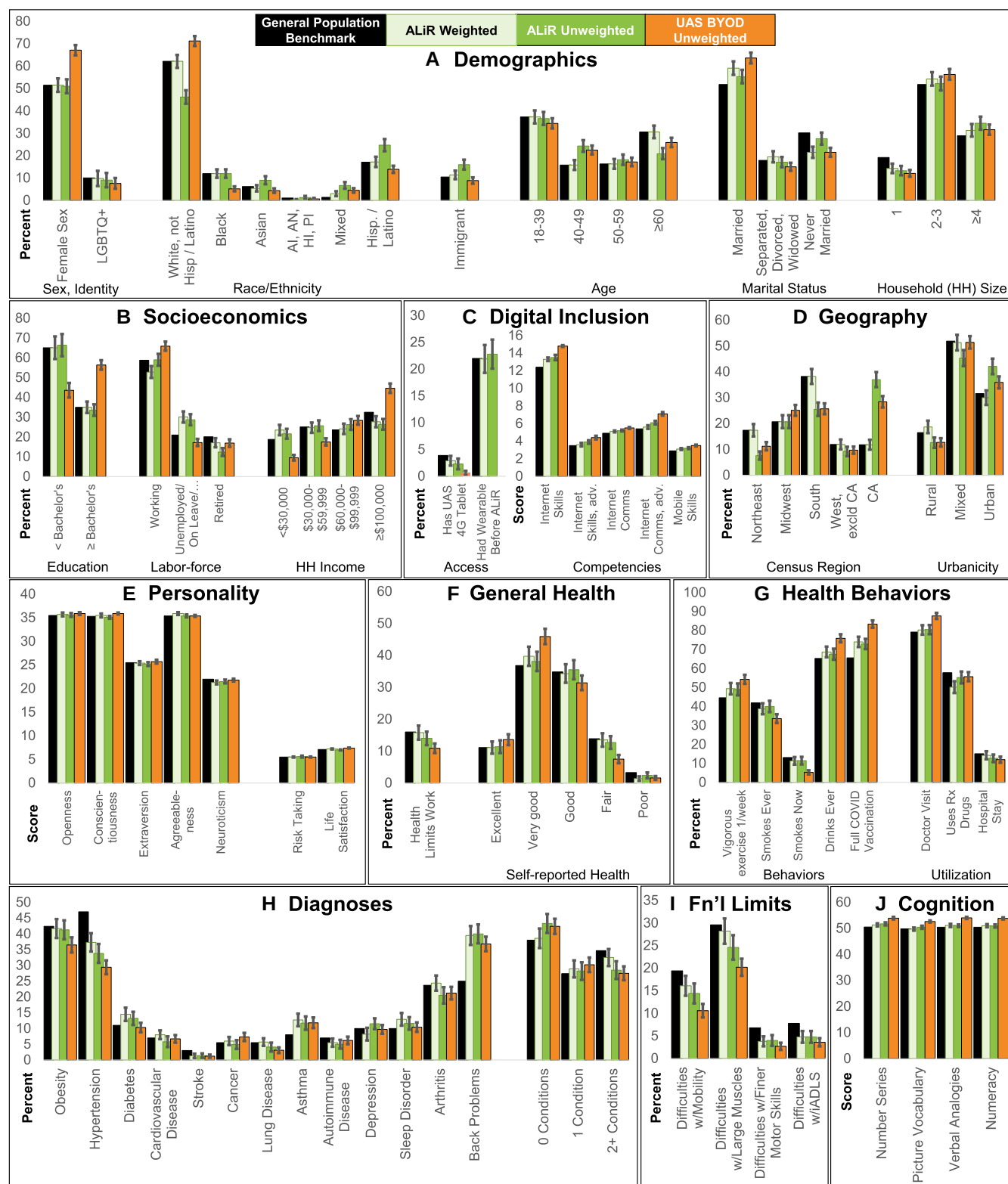


Fig. 4. ALiR characteristics: differences between US adult population norms (black), weighted ALiR (light green), unweighted ALiR (dark green), and UAS-BYOD (orange) across (A) demographics; (B) socioeconomics; (C) digital inclusion; (D) geography; (E) personality; (F–I) health; and (J) cognition. Error bars = 95% CI. Population norms from Basic Monthly CPS (October 2021–March 2022) where available, literature-based epidemiological estimates where available, and from the UAS Comprehensive File weighted to the CPS for remaining variables. Demographics are from the “My Household” survey (April 2021). HRS health/cognitive outcomes are from the UAS Comprehensive File. Chronic conditions = self-reported diagnoses. AI/AN or HI/PI = American Indian/Alaska Native or Hawaiian/Pacific Islander. Fn'l Limits = Functional Limitations. Rx Drugs = Prescription Drugs. See Tables S8–S11 for details.

Alaska Native, Hawaiian, and Pacific Islander by 36%; (v) Hispanic/Latino by 19%; (vi) older adult by 15%; (vii) without bachelor's degree by 33%; (viii) unemployed by 18%; (ix) with household income <\$30 K by 49%; (x) with household income between \$30 and 60 K by 30%; (xi) without UAS-provided Internet access by 85%; and (xii) living in rural areas by 23%. The UAS-BYOD sample would also significantly underrepresent people with diagnosed chronic conditions (obesity, hypertension, lung disease, and stroke), functional limitations, and with poorer overall health, cognitive function, health utilization, health behaviors, financial literacy, and digital competencies.

AoU's BYOD Fitbit dataset ($n = 14,133$) underrepresented males by 41%, non-White participants by 66%, individuals without bachelor's degrees by 57%, and individuals with household incomes <\$75 K by 29% compared to population norms.

Generalizability of models trained on unweighted ALiR vs. BYOD

To illustrate ALiR's generalizability compared with AoU-Fitbit BYOD, respective classification models of COVID-19 infection were trained on each (unweighted) dataset (Fig. 5A; [Online Methods](#)). COVID-19 was chosen due to: (i) comparability of available variables across datasets; (ii) model robustness in the evidence base (3–8); (iii) timeliness and relevance of respiratory infectious disease to the general population; and (iv) known symptom heterogeneities and racial/ethnic disparities in outcomes (6, 31, 32). Drawing on published models (3–8), identical methods were used for data curation, training, and hyperparameter tuning. XGBoost was chosen as the best performing model (area under the curve; AUC) for both datasets. Performance was compared in (ALiR model on ALiR holdout data; AoU on AoU) and out of sample (ALiR model on AoU data, AoU model on ALiR data), overall, and for subgroups defined by intersections of race/ethnicity, age, and sex.

During curation, ALiR's COVID-19 dataset retained 71% of its initial sample vs. AoU's 7%. Demographic distributions, representative for ALiR vs. primarily female, higher-educated, and White for AoU (Fig. 5B; Table S13), remained largely consistent between initial and training samples for both datasets. The prevalence of COVID-19 cases was higher in ALiR (27 vs. 8%), leading to comparable positive training samples ($n = 181$ vs. 237) despite the 14× difference in initial sample size.

ALiR-trained classification performed equally well when tested in (AUC = 0.84, 95% CI: 0.79–0.89) and out of sample, both overall (AUC = 0.84, 95% CI: 0.78–0.89; Fig. 5C) and for subgroups (AUC = 0.82–1.0; Fig. 5D). Conversely, while overall AoU-trained classification performed marginally better in sample (AUC = 0.93, 95% CI: 0.91–0.96), it underperformed by 35% (vs. in sample) when tested on ALiR (AUC = 0.68, 95% CI: 0.61–0.75). Further, AoU's performance was worse in historically underserved groups (Table S14), including older White females (by 40%, despite also having the largest representation in AoU's training dataset) and non-White individuals (by 22–37%) compared with White males (by 17–20%) and younger White females (by 14%).

Discussion

To the best of our knowledge, ALiR is the first longitudinal population study that combines wearable PGHD with comprehensive, frequently repeated, validated labels, meeting all relevant benchmarking criteria for precision health research (15, 16, 36, 62–64). With only two observed sources of differential enrollment (identified from 5,000+ variables), ALiR's 1,038 participants were broadly

representative across demographic, socioeconomic, and health factors, a significant improvement in inclusivity over convenience samples. ALiR's illustrative COVID-19 model (AUC = 0.84) matched or exceeded in-sample performance of others (AUC = 0.52–0.92) (3–8) and maintained effectiveness when tested out of sample and in diverse sociodemographic groups. ALiR's model used a similar number of positive COVID-19 training cases as AoU and other models despite having a 10–100× smaller initial sample (3–8), due to higher (and more consistent with population estimates (65)) disease prevalence and curation-related case retention. In contrast, AoU's BYOD model significantly underperformed on ALiR data, with 22–40% worse performance in older female and non-White groups. Surprisingly, AoU's performance was weakest in its largest training subgroup, likely due to diminished diversity of healthcare-engaged device owners relative to the within-group general population on other dimensions (e.g. education, income, comorbidity, and vaccination) (66). These results suggest that for promoting model generalizability, smaller samples of high-quality, representative data can outperform large convenience samples, as the latter, due to selection effects, may fail to adequately capture the diversity of observed and unobserved characteristics in the target population (67–69). While COVID-19 represents a single conceptual demonstration, its significance as an infectious pandemic that transcended worldwide sociodemographic boundaries is especially pertinent. ALiR's ability to account for heterogeneity in COVID-19 dynamics showcases its potential for improving equity in health applications that have previously suffered from disparities (6, 31, 32). The focus on COVID-19 was further dictated by the availability of analogous data, underscoring the field-wide need for more comprehensive datasets with standardized variables to explore drivers of equitable model performance in different health applications. To that end, probability sampling, hardware provisioning, and strategic oversampling were each critical to ALiR's benchmarking ability, providing a roadmap for inclusive prospective data collection that enhances model generalizability.

Probability sampling of US addresses promoted broad, balanced representation in ALiR's invited sample and minimized expected sources of selection bias (17, 39). In contrast, studies that recruit from clinical encounters (e.g. UK Biobank, AoU) are more likely to underrepresent people experiencing healthcare access barriers, who also tend to be non-White, under-resourced, and/or of poorer health (17–20, 70, 71). Further, probability sampling likely better accounted for unobserved factors (66, 72, 73), leading to ALiR's stronger performance on nonrepresentative data and AoU's poorer performance on representative data, even within larger subgroups. This suggests that in the context of convenience samples, use of subgroup targets and geography-based recruitment to achieve cohort inclusivity, as in AoU's sampling approach (18), may be insufficient for model generalizability.

Universal Fitbit provisioning enabled invited individuals to remain eligible regardless of self-motivated technology access, which minimized differential enrollment and preserved inclusivity of the probability sample. ALiR's experience suggested that 64% of invited individuals were willing to participate, though only 23% of enrollees owned a wearable at the time, consistent with a recent study suggesting cost and device-related education gaps explained a similar discrepancy in federally qualified health center patients (58 vs. 20%) (25). It is not surprising then, that BYOD designs have systematically underrepresented populations experiencing health disparities (21). Indeed, had ALiR relied on UAS device owners, the invited sample would have substantially underrepresented people of color, older adults (22–25), those

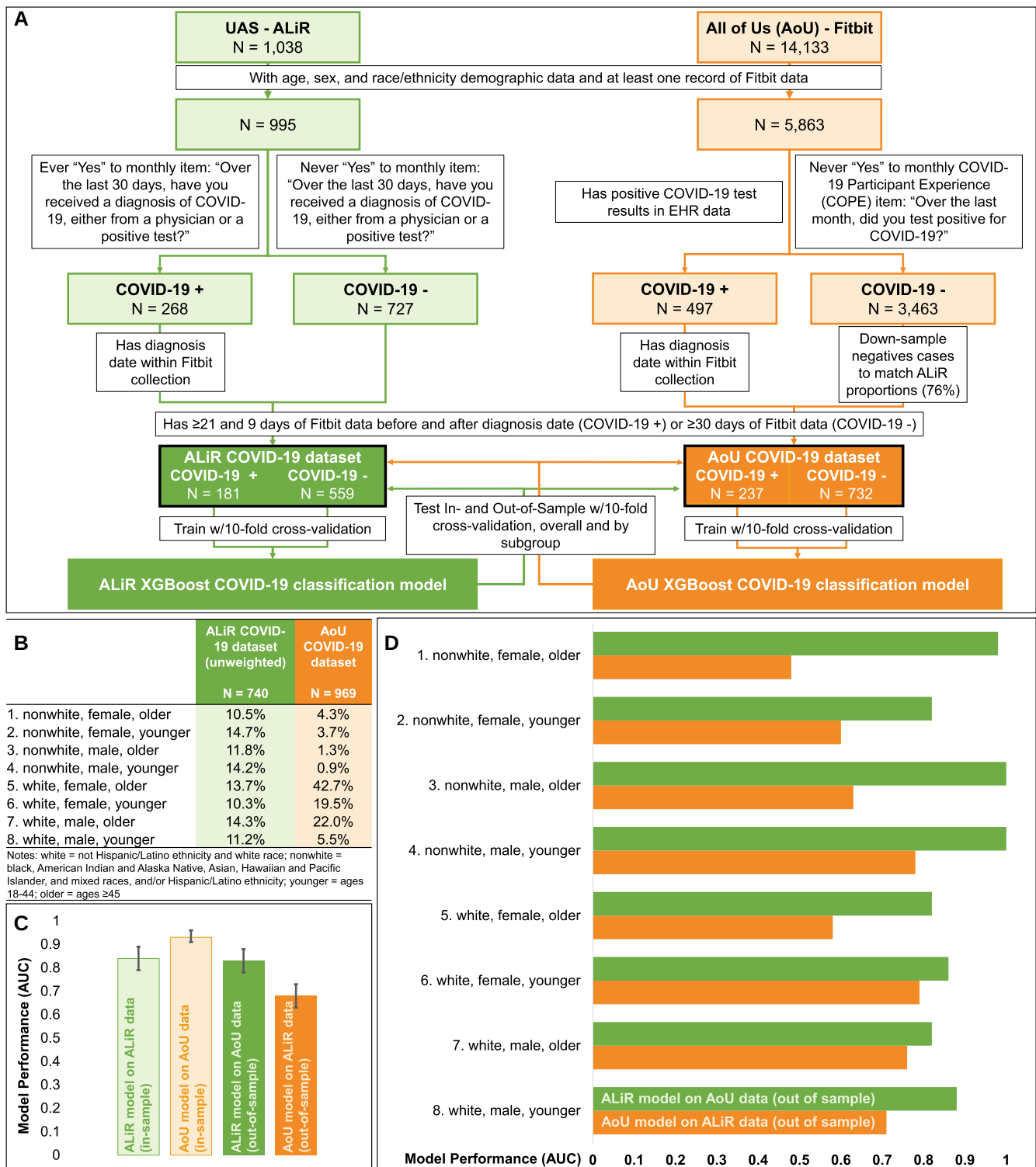


Fig. 5. Generalizability of COVID-19 classification model performance using ALiR vs. AoU datasets: A) Schematic representation of COVID-19 classification model training and testing procedure for ALiR (green) and AoU (orange) datasets. COVID-19-positive cases = COVID+; COVID-19-negative cases = COVID-. B) Sample composition of ALiR (unweighted) and AoU datasets. C) Overall COVID-19 classification model performance (area under curve; AUC) in-sample (ALiR-trained on ALiR holdout data: light green and AoU-trained on AoU data: light orange) and out of sample (ALiR-trained on AoU data: dark green and AoU-trained on ALiR data: dark orange). D) Out-of-sample COVID-19 classification model performance (AUC) for sociodemographic subgroups for ALiR-trained model tested on AoU data (green) and AoU-trained model tested on ALiR data (orange).

with lower income or education, and those with poorer internet literacy, financial literacy, health, and cognitive ability—disparities that would likely grow along age, education, and associated dimensions during enrollment. AoU's BYOD Fitbit cohort was

even less representative, highlighting the compounding effect of combining convenience sampling methods.

Strategic oversampling, on one hand, led to overrepresentation of racial/ethnic minorities and socioeconomically disadvantaged

individuals—who did not exhibit lower consent or enrollment compared with their counterparts—in ALiR's unweighted cohort. On the other, oversampling counterbalanced relatively poorer enrollment rates in individuals with lower education who were equally willing to participate but faced barriers beyond device access after consenting. Address confirmation and recurring unavailability to sign for packages were the top reasons for loss to follow-up, which may reflect greater housing instability or time pressures within this population (74). Remote recruitment is a primary strategy in large-scale PGHD research, and planned investigations will elucidate how enrollment steps may inadvertently introduce additional barriers in underserved populations.

Older adults were the only underserved or disparity-facing group also underrepresented in ALiR's unweighted cohort as we did not anticipate their relative unwillingness to participate despite Fitbit provisioning. This disparity, apparent even after controlling for many other age-related factors (employment, income, cognitive ability, chronic disease, disability, Internet skills, etc.), is consistent with the BYOD literature (22). Again, barriers beyond device access are likely involved, including mistrust of, disinterest in, or lack of perceived benefit from using wearables or participating in wearable studies (33–35). Facing a globally aging population, precision health offers an important opportunity to support older adult health and well-being (e.g. PGHD-based detection of depression (75), cognitive decline and dementia (76, 77), and falls (78)), but systematically improving their research participation must be a priority. As we grow ALiR's cohort to ~10,000 members by 2027 alongside UAS's ongoing expansion, we will continue to characterize and quantify the mechanisms driving differential study participation to maintain and refine inclusivity. Potential interventions include: (i) oversampling older adults in future recruitment; (ii) strategically inviting preexisting UAS device owners; and (iii) employing evidence-based strategies like community-based participation efforts that improve trust, education, and engagement in minoritized and/or underrepresented communities (79–81).

This study is not without limitations. First, we focus on consent and enrollment; longitudinal engagement—the third component of participation—is the subject of ongoing work. Second, while ALiR substantially improves representativeness relative to existing PGHD cohorts, some selection biases remain. Weighting did not fully correct underrepresentation among older adults not in the labor force (e.g. retired, unemployed, on leave, or disabled), highlighting the potential for residual bias from unobserved factors, despite extensive coverage of observed characteristics. Third, while our illustrative COVID-19 model demonstrates promise for improved generalizability, it represents a single example. Future work will assess model performance across additional health domains.

Collectively, ALiR provides the scientific community with a public benchmarking toolkit: a methodological framework and connected cohort for prospective research, and an inclusive dataset for training and validating generalizable precision health models in diverse populations.

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Supplementary Material

Supplementary material is available at [PNAS Nexus](#) online.

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Data Availability

As UAS makes deidentified study data available to registered researchers (who sign the Data Use Agreement), ALiR PGHD will be available upon curation of one full year of data for the entire panel (anticipated in late 2025) and will be updated frequently thereafter. This will include additional study information, such as data dictionaries, protocols, informed consent documentation, and app code. COVID modeling code developed for this study is also available. All above information can be obtained from the UAS website at <https://uasdata.usc.edu/>. A portion of this work relies on data from the All of Us Research Program; data and relevant analysis code are available to select US universities on the Researcher Workbench (<https://www.researchallofus.org/data-tools/workbench/>).

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